IoT Motion Detector with Email Alert

INTRODUCTION

The IoT Motion Detector with Email Alert is a smart security monitoring system that detects human movement using a PIR sensor and sends an alert notification through email using the internet. The project uses the ESP8266 NodeMCU Wi-Fi module to connect with IFTTT Webhooks and automatically trigger email alerts whenever motion is detected. This system is useful for home security, office monitoring, and smart surveillance applications.

OBJECTIVE

- To detect motion using a PIR sensor.
- To send automatic email alerts when motion is detected.
- To understand IoT communication using ESP8266 Wi-Fi.
- To integrate IFTTT Webhooks with hardware devices.
- To create a low-cost smart security system...

COMPONENTS REQUIREMENTS

ESP8266 NodeMCU

LED + 220 ohm Resistor

Breadboard

Jumper Wires

PIR Sensor (HC-SR501)

USB Cable

Wi-Fi Connection

Android Mobile Phone

WORKING PRINCIPLE

The IoT Motion Detector with Email Alert works by using a PIR (Passive Infrared) sensor to detect movement in its surrounding area. The PIR sensor senses changes in infrared radiation produced by humans or moving objects and sends a signal to the ESP8266 NodeMCU whenever motion is detected. The NodeMCU processes this signal and turns ON an LED to indicate motion detection. At the same time, it connects to the internet through Wi-Fi and sends a request to the IFTTT Webhooks service. IFTTT then automatically triggers an email notification to the user, informing them about the detected motion. A delay or cooldown time is included in the program to prevent continuous repeated alerts.

SOFTWARE USED

- Arduino IDE – Used to write and upload code to ESP8266.
- IFTTT – Used for email automation and notifications.
- ESP8266 Board Package – Enables NodeMCU programming in Arduino IDE.

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APPLICATIONS

- Home security systems
- Intruder detection systems
- Smart office monitoring

CODE

```
#include <ESP8266WiFi.h>
#include <ESP8266HTTPClient.h>
#include <WiFiClientSecure.h>

const char* ssid = "V2022";
const char* password = "Sheetal2004";
const char* iftttKey = "dAbC123XyZ456";
const char* event = "motion_detected";

#define PIR_PIN D2
#define LED_PIN D1

unsigned long lastAlert = 0;
const unsigned long cooldown = 30000; // 30 second cooldown

void sendAlert() {

  WiFiClientSecure client;
  client.setInsecure(); // Skip SSL verification for demo

  HTTPClient http;

  String url =
    "https://maker.ifttt.com/trigger/" +
```

```
    String(event) +
    "/with/key/" +
    String(iftttKey);

  http.begin(client, url);

  http.addHeader("Content-Type", "application/json");

  int code = http.POST("{\"value1\":\"Motion detected!\"}");

  Serial.println("IFTTT response: " + String(code));

  http.end();
}

void setup() {

  Serial.begin(115200);

  pinMode(PIR_PIN, INPUT);
  pinMode(LED_PIN, OUTPUT);

  WiFi.begin(ssid, password);

  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }
```

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```
  Serial.println("\nWi-Fi connected. System armed.");
}

void loop() {

  if (digitalRead(PIR_PIN) == HIGH) {

    digitalWrite(LED_PIN, HIGH);

    if (millis() - lastAlert > cooldown) {

      Serial.println("Motion detected! Sending alert...");

      sendAlert();

      lastAlert = millis();
    }

  } else {

    digitalWrite(LED_PIN, LOW);
  }

  delay(200);
}
```